

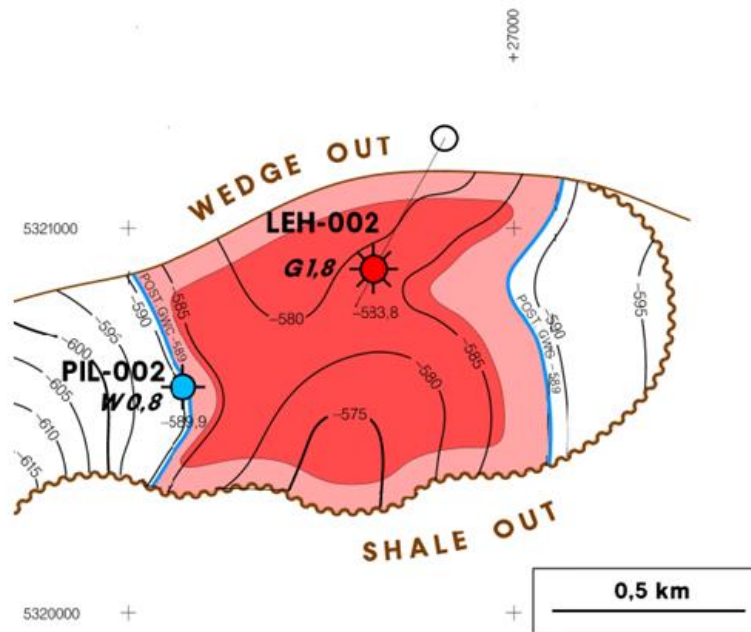
Hydrogen storage and geo-methanation in a depleted underground hydrocarbon reservoir

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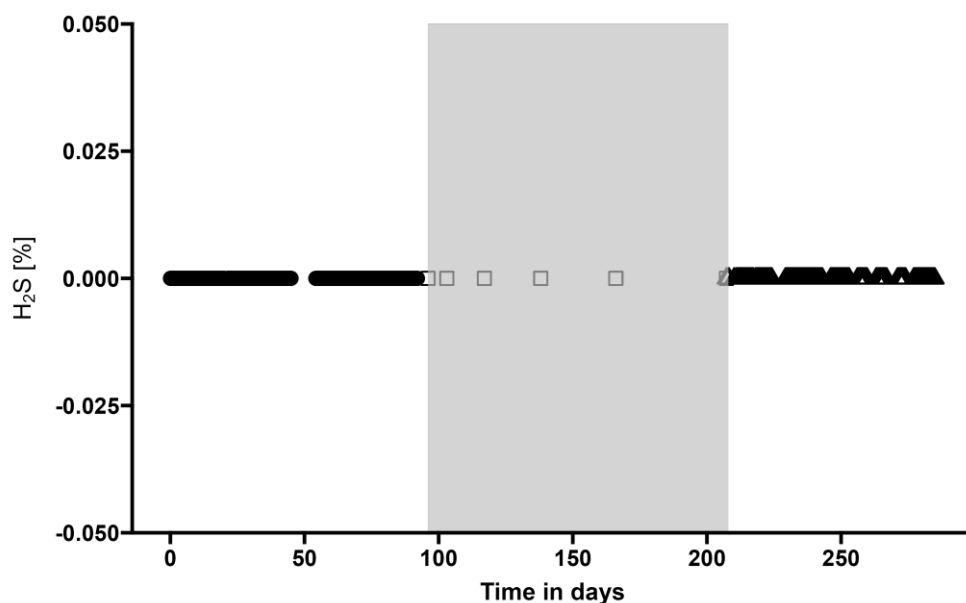
Supplementary Information

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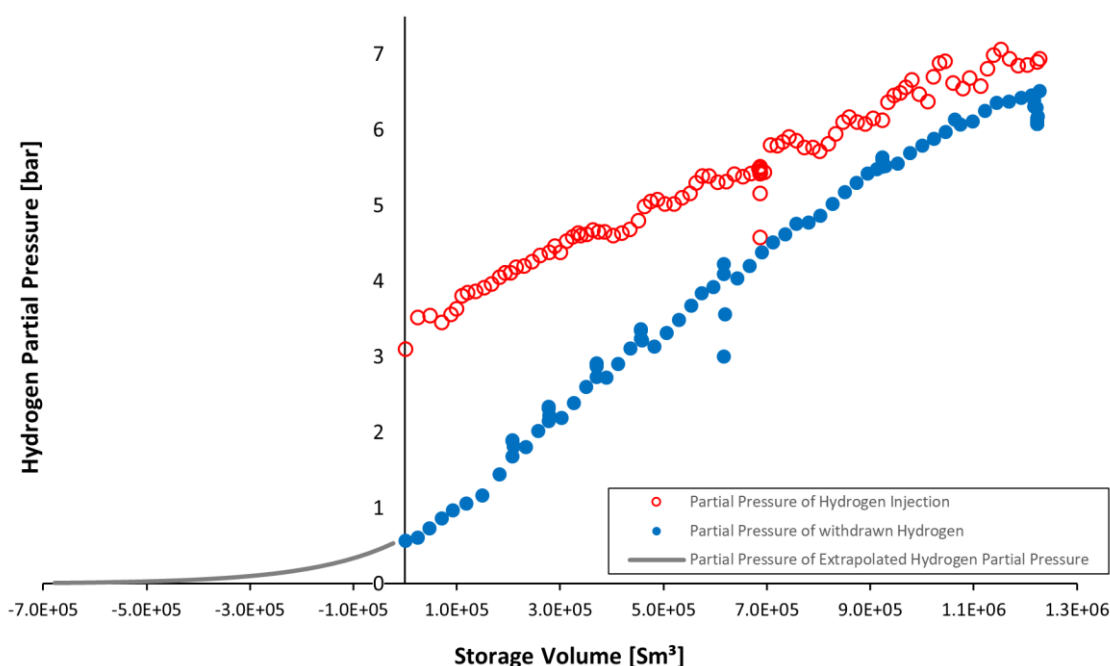
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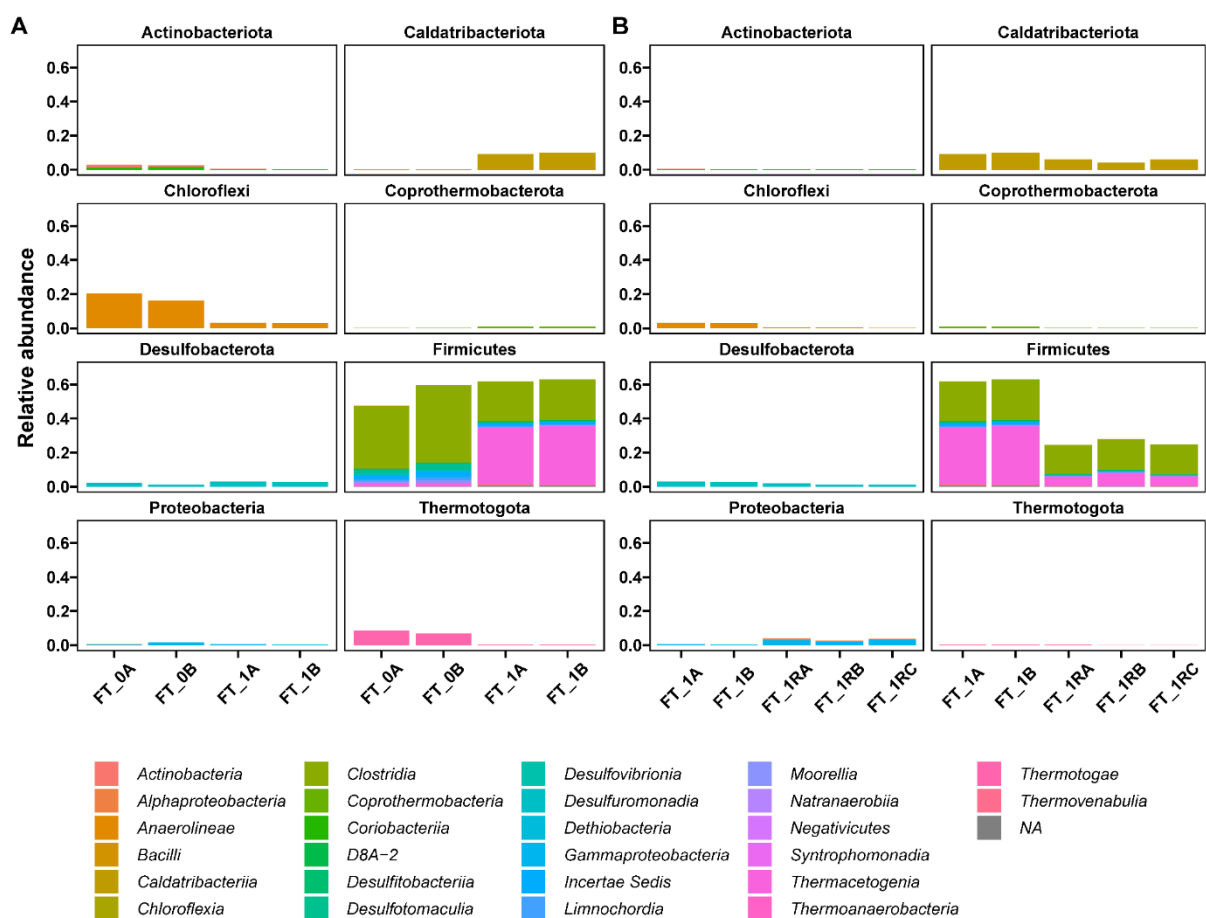
Supplementary Figure 1: Structural Map of the test reservoir Lehen. The map shows the seismic derived shape of the reservoir with the dark red area representing the confirmed gas bearing zone and the light red area the expected gas bearing zone. The gas-water contact is indicated by blue lines. The z coordinates are referenced to sea level and not to ground level. The well PIL-002 displayed in the map was sealed before the start of the field trial. The captions “Wedge Out” and “Shale Out” define the boundaries of the reservoir with the northern wedge out being a tectonic boundary and the southern shale out being a sedimentary boundary.



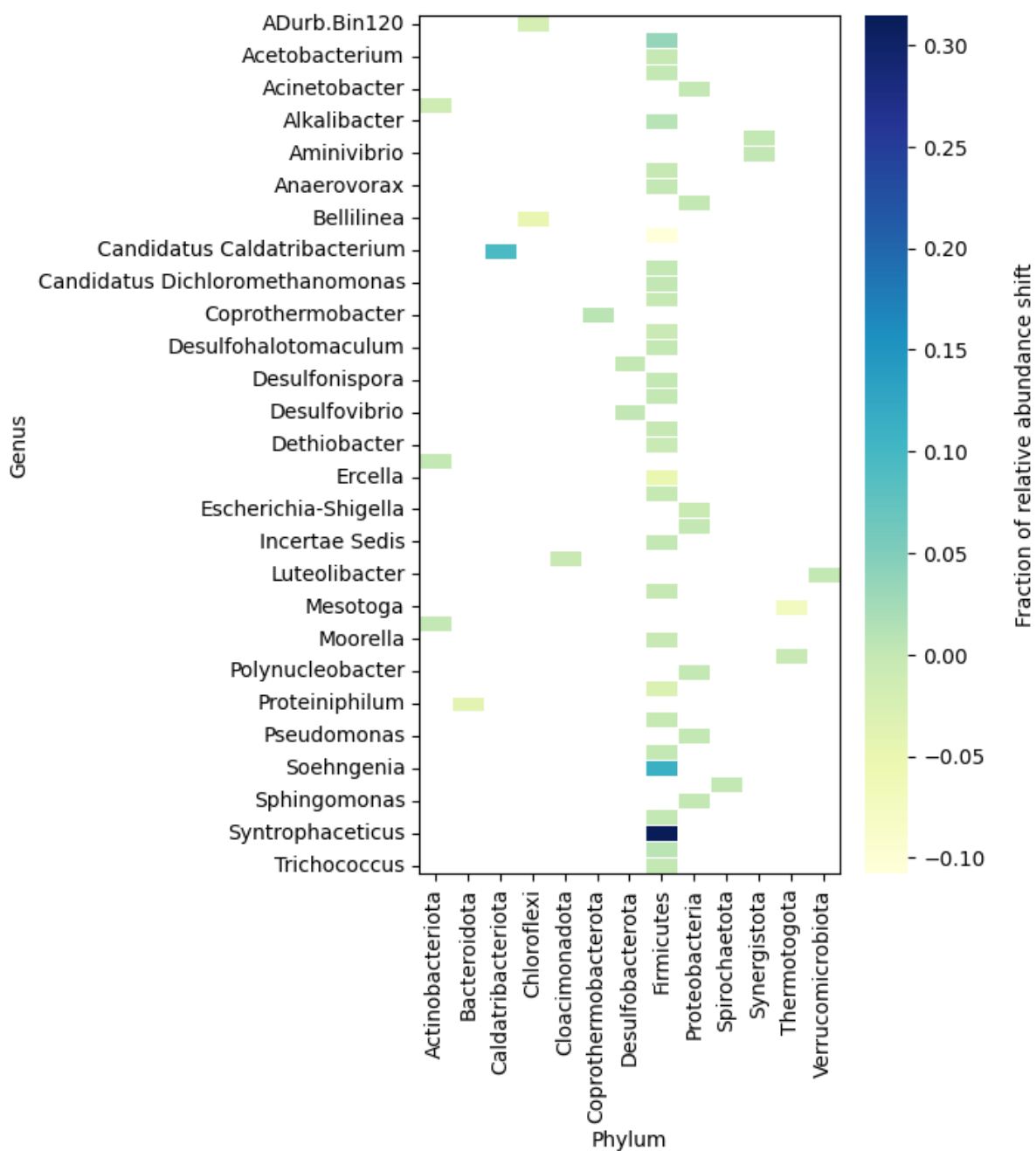
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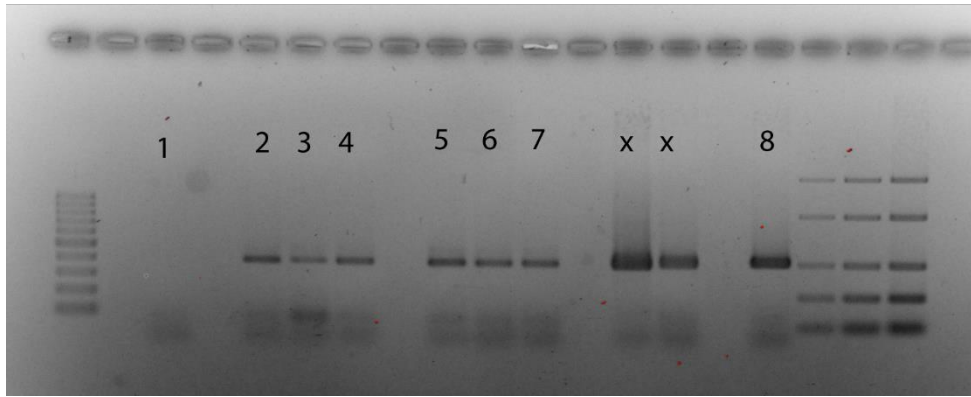
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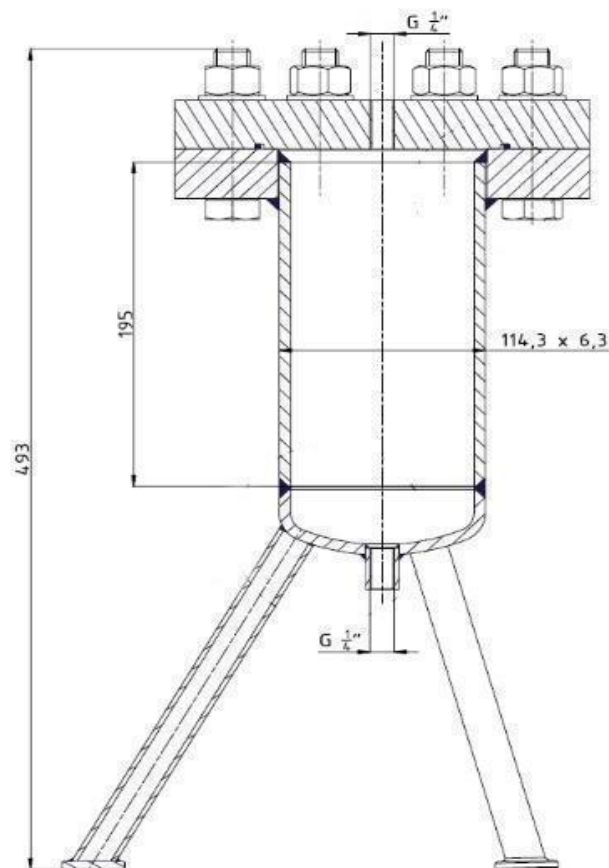
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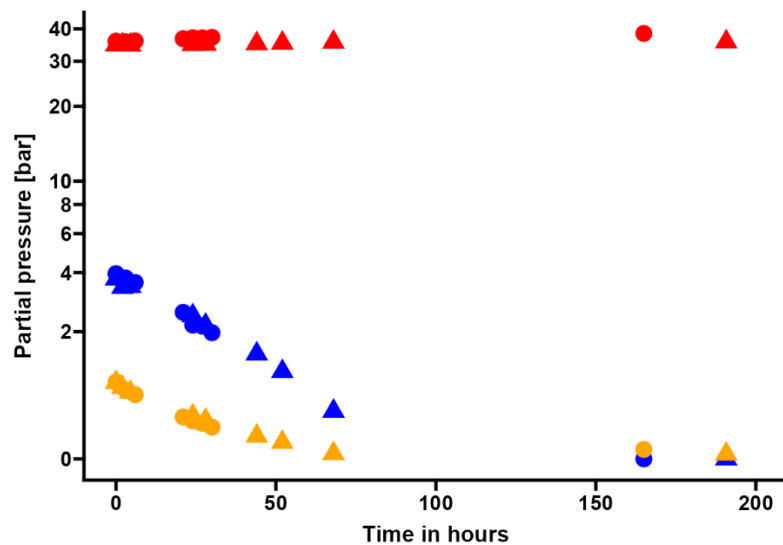
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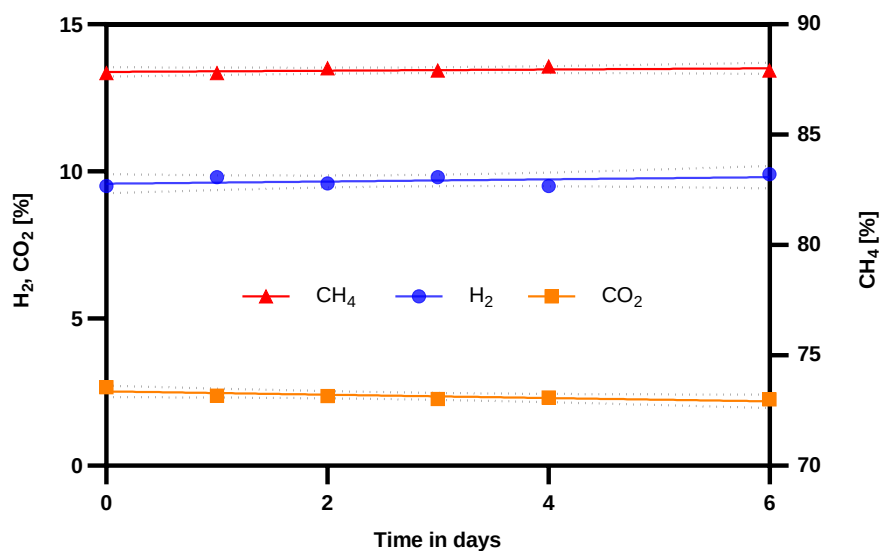
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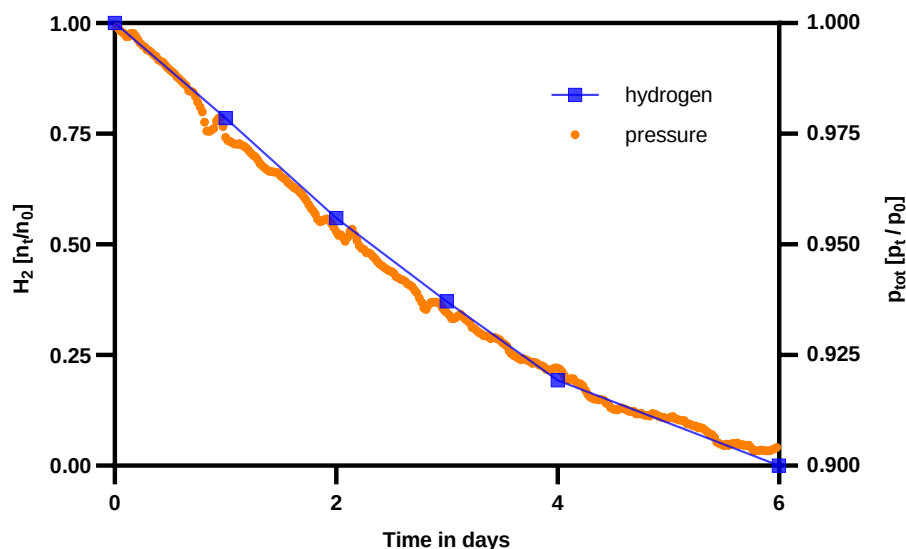
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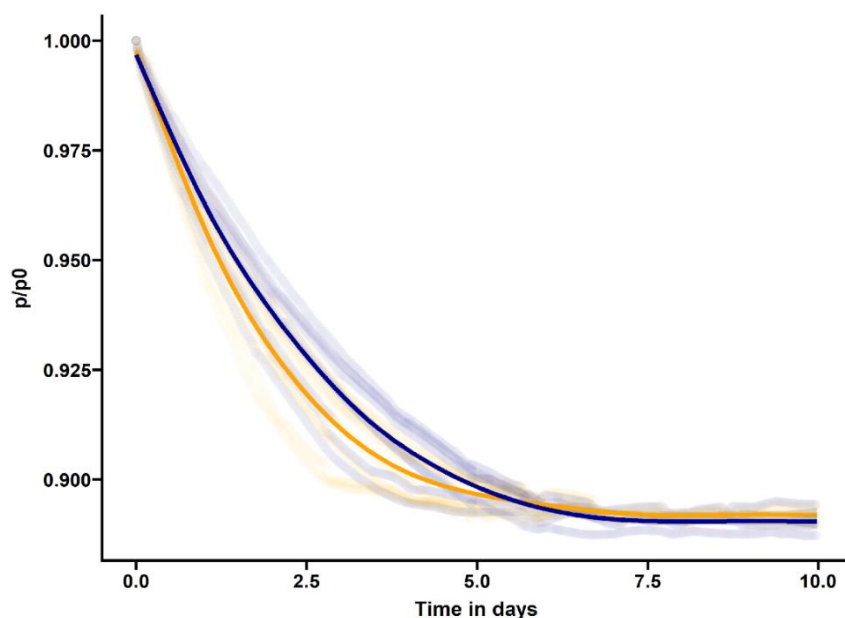
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Supplementary Figure 9: Headspace gas concentrations in abiotic control during incubation with substrate gas mixture. Concentration of individual gases (% (v/v)) in head space of abiotic mesocosm over a period of 6 days (CH₄ – red triangles, H₂ – blue circles, CO₂ – orange squares). Full lines designate linear regression of data sets, dotted lines indicate confidence intervals (95 %). Slope of linearly regressed concentration data does not differ significantly from zero for H₂ (p-value = 0.352) and CH₄ (p-value = 0.311).



Supplementary Figure 10: Gas conversion and decrease of mesocosm pressure over time in biotic mesocosm. Corrected and normalized total mesocosm pressure (p_{tot}) and mole fraction of H_2 during one geo-methanation cycle in a biotic reservoir mimicking mesocosm (M6, see Methods) supplied with substrate gas mix containing 10 % (v/v) H_2 and 2.5 % (v/v) CO_2 to an absolute pressure of 44.3 bar. Loading of M6 with sandstone core and reservoir brine was conducted in the same manner as for M3. Before this representative cycle, 12 conversion cycles with above-mentioned substrate gas mix had been performed in M6. p_0 : initial pressure, n_0 : amount of H_2 in moles introduced into the mesocosm at the beginning of the cycle. Pressure correction and normalization procedures are detailed in the Methods section.



Supplementary Figure 11: General additive models (GAM) of normalized pressure development due to substrate gas conversion in high pressure mesocosms M1 (orange, $n=10$, K_2HPO_4 added) and M2 (dark blue, $n=10$, no supplement). Opaque symbols represent individual data points of M1 (orange) and M2 (blue).

Supplementary Table 1: Composition of carrier gas (working gas) used for co-injection with hydrogen into the Lehen reservoir. Gas components present in traces are not listed.

Gas component	Concentration [% (v/v)]
Nitrogen	0.53
Carbon Dioxide	0.22
Methane	98.6
Ethane	0.44
Propane	0.07
Iso-Butan	0.05
n-Butan	0.02
Iso-Pentane	0.03
n-Pentane	0.01

Supplementary Table 2: Hydrogen gas volumes operated in the field trial.

Phase	Volume [Sm ³]
Injection	118196
Production	99754

Supplementary Table 3: Composition of cushion gas initially present in the Lehen reservoir. Gas components present in traces are not listed.

Gas component	Concentration [% (v/v)]
Nitrogen	0.54
Carbon Dioxide	0.19
Methane	98.4
Ethane	0.49
Propane	0.18
Iso-Butan	0.03
n-Butan	0.03
Iso-Pentane	0.07
n-Pentane	0.01

Supplementary Table 4: Acetate concentrations in brine of non-H₂ exposed depleted gas reservoirs quantified via ion chromatography (Methods). All sites are located within a radius of 100 km and situated in the Molasse Basin. Reservoir brine sampling from all wells was conducted as described in the Methods section.

Sampling date	Reservoir name	Sampling depth [m]	pH	Acetate [mg L ⁻¹]
07.03.2017	Bamberg	1014	6.6	148
08.03.2017*	Lehen	1195	7.9	102
21.03.2018	Nussdorf	1500	8.0	< LOQ
10.04.2019	Nussdorf	1500	7.8	106
10.10.2018	Rubensdorf	1650	8.8	768
27.02.2020	Rubensdorf	1220	7.9	589

*) First sampling event after completion of field trial (10.01.2017)

Supplementary Table 5: CO₂ balance of field trial

CO ₂ [m ³]	Phase of storage cycle
2297.8	Injection
190.3	Cushion gas
1523.8	Production
3.8	Shut-in
960.5	Difference

Supplementary Table 6: Development of gas concentrations in the test reservoir after completion of the field trial. Number of days denotes elapsed time after end of the field trial.

Days	H ₂ [% (v/v)]	CO ₂ [% (v/v)]	CH ₄ [% (v/v)]
0	2.82	0.07	96.7
42	1.46	0.04	98.1
174	1.14	0.01	97.9
580	0.95	0.03	99.0

Supplementary Table 7: XRD analysis of drill cores originating from the thrustured Puchkirchen formation of the Sierning natural gas reservoir. This formation is part of the Oligocene Epoch within the Tertiary Period. The specific reservoir was deposited during the Egerian Phase (20.8 -27.5 Mio years ago).

Minerals	[%]
Quartz	36
Potassium feldspar	3
Plagioclase	7
Calcite	8
Dolomite	20
Siderite	1
Clay minerals and mica	25

Supplementary Table 8: Elemental analysis of reservoir brine and sedimented particulates.

Sample	Ca	Co	Cu	Fe	K	Mg	Mn	Mo	Ni	P	S	Zn	TN ^a
Reservoir brine ^b [mg L ⁻¹]	27.7	<LOD ^c	<LOD	34.9	7130	25.6	0.485	0.079	0.056	0.133	3.15	0.123	8
Brine sediment [mg kg ⁻¹]	17900	10.3	125	64300	8390	2190	1080	60.7	255	407	9570	258	NA ^d

^{a)} total nitrogen

^{b)} untreated brine sampled at 1200 m, including suspended solids used to inoculate mesocosms.

^{c)} LOD: limit of detection

^{d)} NA: not available

Supplementary Table 9: Chemical analysis of reservoir brine and liquid of biotic mesocosm.

Sample	Days ^a	Cl ⁻ [mg L ⁻¹]	NO ₃ ⁻ [mg L ⁻¹]	PO ₄ ⁻³ [mg L ⁻¹]	SO ₄ ⁻² [mg L ⁻¹]	Acetate [mg L ⁻¹]	pH
filtered reservoir brine ^b 1200 m	-13	7487	11.5	<LOD ^c	<LOQ ^d	206	8.7
Mesocosm 1	0	6613	<LOD	<LOQ	<LOQ	276	7.3
Mesocosm 2	0	6795	<LOD	<LOQ	<LOQ	274	7.3
Mesocosm 3	0	6597	<LOD	<LOD	<LOD	<LOQ	7.1
Mesocosm 1	70	NA ^e	NA	NA	NA	<LOD	7.1
Mesocosm 2	70	NA	NA	NA	NA	<LOD	7.1
Mesocosm 3	52	6448	<LOD	<LOD	<LOD	<LOQ	7.0
Mesocosm 1	92	NA	NA	NA	NA	<LOD	6.6
Mesocosm 2	92	7065	<LOQ	<LOQ	<LOQ	<LOD	6.9
Mesocosm 3	344	5936	<LOD	<LOD	<LOD	112	6.9
Mesocosm 6	176	6656	<LOD	<LOD	NA	53	7.1

^{a)} time in days relative to start of mesocosm experiments

^{b)} brine including suspended solids used to inoculate mesocosm

^{c)} LOD: limit of detection

^{d)} LOQ: limit of quantification

^{e)} NA: not available

Supplementary Table 10: List of oligonucleotides used in this study.

Primer Name	Sequence 5'- 3'
515_F	GTG CCA GCM GCC GCG GTA A
806_R	GGA CTA CHV GGG TWT CTA AT
NGS_515_F	TCG TCG TCA GCG TCA GAT GTG TAT AAG AGA CAG GGT GCC AGC MGC CGC GGT AA
NGS_806_R	GTC TCG TGG GCT CGG AGA TGT GTA TAA GAG ACA G GGA CTA CHV GGG TWT CTA AT
iMeA_1046_F	TAY ATG WSI GGH GGI GTI GGI TTY AC
iMeA_1435_R	TGR TCY TGI ARR TCR WAI CCR WAG AAI CC
NGS_iMeA_1046_F	TCG TCG GCA GCG TCA GAT GTG TAT AAG AGA CAG TAY ATG WSI GGH GGI GTI GGI TTY AC
NGS_iMeA_1435_R	GTC TCG TGG GCT CGG AGA TGT GTA TAA GAG ACA GTG RTC YTG IAR RTC RWA ICC RWA GAA ICC

Supplementary Note 1: Sand production measurements

The sand level in the well was unchanged (1424.5 m total vertical depth) from 1st of March 2016 to the 25th of July 2019 and 1st of February 2021.

Supplementary Note 2: Diffusion and dispersion of gases in porous reservoir rock

Gas migration in porous rock formations is affected by diffusive and dispersive forces. Diffusion is a process by which an inhomogeneous, multicomponent gaseous system on which no work is accomplished will strive towards dynamic equilibrium. In porous media, the movement of gases is restricted, which negatively impacts diffusivity or the diffusion coefficient of a gas (D) and is considered in the effective diffusion coefficient (D_{eff} , Supplementary equation 1). The porosity (ϕ) of the formation denotes the ratio of pore to bulk volume and tortuosity (τ). It is a measure of how strongly the flow path in the porous medium diverges from a straight, cylindrical tube. Supplementary equation 2 describes the diffusive flux of gases in a binary system, where J_{diff}^k is the diffusive flux of component k in $\text{mol m}^{-2} \text{s}^{-1}$, ρ is the molar density of gas in $\text{mol}^{-1} \text{m}^{-3}$, D_{eff} is the binary effective diffusion coefficient in $\text{m}^2 \text{s}^{-1}$ and ∇x^k is the gradient of the mole fraction of component k . From Supplementary equation 2 it follows that the diffusive flux of gases in a porous medium is a function of the concentration gradient of the gaseous component. In addition, the presence of other fluids in the pores lowers the diffusivity of the gases and prolongs the process of equilibration².

Mechanical dispersion describes a mixing process induced by applying an external force. It follows from Supplementary equation 3 (where J_{disp}^k is the dispersive flux of component k in $\text{mol m}^{-2} \text{s}^{-1}$, ρ is the molar density of gas in mol m^{-3} , α is the mechanical dispersivity in m, U is the true flow velocity in m s^{-1} and ∇x^k is the gradient of the mole fraction of component k), that mechanical dispersion is directly proportional to the flow velocity of the gas in the porous medium. In the porous reservoir, the movement of gas molecules is induced through the increase of pressure during gas injection. Furthermore, the mobility of the mixing fluids and inherent porous medium parameters (i.e., porosity, tortuosity, and pore throat size)³ influence dispersion.

$$D_{eff} = \frac{D\phi}{\tau} \text{ (Supplementary equation 1)}$$

$$J_{diff}^k = -\rho D_{eff} \nabla x^k \text{ (Supplementary equation 2)}$$

$$J_{disp}^k = -\rho \alpha U \nabla x^k \text{ (Supplementary equation 3)}$$

Supplementary Note 3: Mesocosm instrumentation

Pressure and temperature of all mesocosms were monitored at 15 to 30-minute intervals using digital PA-33 X sensors (KELLER AG für Druckmesstechnik; pressure range 0 to 100 bar, temperature range -10 °C to 80 °C) mounted to the upper shell of the mesocosm (G ¼" port).

For gas transfer, a needle valve (Hoke® 1700 Series, # 1751G6YMM) was mounted to the upper shell (G ¼" port) of the mesocosm using Hoke® Gyrolok adapters. For withdrawal of liquid from the mesocosm, a ball valve (Hoke® Flomite 71 Series, # 7122G6YMM) was connected to the bottom (G ¼" port) of the mesocosm using Hoke® Gyrolok adapters. The mesocosms were equipped with a burst disc (type WWS_SUM-4G, SCHWING Verfahrenstechnik GmbH) featuring a burst pressure of 65 bar at 47 °C. Burst discs were fixed to the upper shell of the mesocosm (G ½" port) and connected to a safety tubing system for gas release to the exterior. Heating sleeves (Carlo Loysch GmbH; Mikanitringheizbänder, diameter 114x120 mm, 50W, 230V) were attached to the cylindrical part of the mesocosms, which were then insulated with customized mantles made from expanding foam (B1 Soudafoam, Soudal N.V. Germany).

Digital instrumentation was set up in-house for mesocosm pressure monitoring and temperature control, comprising a CompactLogix, 16DI, 16DO, 24VDC PS, and CompactLogix ASCII Interface Modul (Routeco GesmbH). The PLC (programmable logic controller) was coded in "RS Logix" whereas data logging and visualization were accomplished via "FactoryTalk" (Rockwell Automation) software.

Supplementary Note 4: Chemical analysis of reservoir brine and mesocosm liquids

Anion species Cl^- , NO_3^- , SO_4^{2-} and PO_4^{3-} were quantified in liquid samples using ion chromatography (Dionex ICS-900, equipped with an AS-DV auto-sampler, an ACRS-500 suppressor and a DS5 detector). Analytes were separated using an Ionpac TM AS14A analytical column (4 x 250 mm) equipped with an AG14A guard column (4 x 50 mm). Before injection, samples were filtered using 0.45 μm filters (CA-45/15MS, MACHERY-NAGEL Chromafil). The instrument was operated in suppressed conductivity detection mode at isocratic conditions with a flow rate of 1 mL min^{-1} . The mobile phase comprised 1 mM NaHCO_3 (Merck KGaA) and 8 mM Na_2CO_3 (Merck KGaA), and was also used for sample dilution. The conductivity suppressor was regenerated with 31.5 mM H_2SO_4 (Merck KGaA). Samples (diluted 1:10 and 1:100) were measured in triplicates. Chromatograms were integrated automatically. External standard calibration was performed for each analyte.

Supplementary Table 11: Detection and quantification limits for anion species analyzed with ion chromatography.

Anion	LOD [mg L^{-1}]	LOQ [mg L^{-1}]	ULOQ [mg L^{-1}]
Chloride	0.08	0.23	175.4
Nitrate	0.17	0.5	352.3
Phosphate	0.69	2.06	698.8
Sulphate	0.28	0.85	698.8

LOD limit of detection

LOQ limit of quantification

ULOQ upper limit of quantification

Electrical conductivity (EC) and pH of pristine samples were measured using a HACH HQ40d multi-base unit equipped with a HACH IntelliCAL PHC101 standard gel-filled pH electrode and a HACH IntelliCAL CDC401 standard conductivity probe (Hach Lange GmbH).

Total nitrogen content (TN) of filtered (Aqua 30 / 0.45 CA, Merck KGaA) reservoir brine was quantified using a SHIMADZU TOC-V CPH analyzer coupled to a SHIMADZU TNM-1 detector for nitrogen quantification (range 0 – 200 mg L^{-1} , LOD 5 mg L^{-1}).

Elemental analysis of reservoir brine and sediment was conducted by a service provider (atres Group) using ICP-OES (Inductively Coupled Plasma - Optical Emission Spectrometry). Elemental analysis was performed according to DIN ISO 11885. The limit of detection (LOD) of individual elements as quantified in mesocosm liquid and reservoir brine using ICP-OES analysis is given in the following table.

Supplementary Table 12: Detection and quantification limits for elements analyzed via Inductively Coupled Plasma - Optical Emission Spectrometry; LOD - limit of detection.

Element	LOD [mg L ⁻¹]
Calcium	0.014
Cobalt	0.003
Copper	0.099
Iron	0.057
Potassium	2.039
Magnesium	0.002
Manganese	0.003
Molybdenum	0.002
Nickel	0.004
Phosphorus	0.04
Sulphur	0.264
Zinc	0.006

Acetate was quantified by a service provider (atres Group) using GC-FID (gas chromatography - flame ionization detector). An aliquot of 10 mL of each sample was sent to the laboratory for analysis, which was performed according to the company standard HV-01 (LOQ = 55 mg L⁻¹).

Supplementary Note 5: Set-up of gas chromatographic system for the analysis of mesocosm headspace gas samples

An AGILENT gas chromatograph (7890A) was employed for gas analysis. Helium (quality 5.0, Messer Austria GmbH) was used as a mobile phase after an additional purification step (Valco Instruments Co. Inc.). Gas samples were split into separate streams to detect individual components. Stream 1 was equipped with a PlotQ15 Agilent 19095P-Q03 [15 m x 530 µm x 40 µm] and a PlotMS Agilent 19095P-MS6 column [30 m x 530 µm x 50 µm], leading to a pulse

discharge detector (PPD) for quantifying H₂, H₂S, O₂ and N₂. In stream 2, CH₄ and CO₂ were separated on a PlotQ30 Agilent 19095P-Q04 column [30 m x 530 µm x 50 µm] and detected using an FID (CO₂ reduction before detection using H₂ and a nickel catalyst). Each sample was analyzed twice using a split ratio of 1:10 and 1:25, enlarging the calibration range to meet concentrations of all individual gas components (injection volume 2.5 µL, inlet temperature 105 °C). The limits of detection and quantification (both in % (v/v)) are for CH₄ 0.39 and 0.77, for H₂ 0.05 and 0.10, and for CO₂ 0.07 and 0.13, respectively.

Supplementary Note 6: DNA extraction procedure

Reservoir brine or mesocosm liquid (50 mL each) was filtered (SuporR 200, 0.2 µm, Ø 47 mm, PALL Life Sciences) to concentrate biomass. The fragmented filters were dispersed in 400 µL of sterile TE buffer (10 mM Tris-HCl pH 7.5, 1 mM EDTA, Sigma-Aldrich). Cell disruption and membrane disintegration were accomplished by bead beating using glass beads sterilized by autoclaving and by the addition of SDS (sodium dodecyl sulfate, Sigma Aldrich) to a concentration of 3 % (w/v). Thereafter, the samples were homogenized for 30 seconds using a vortex mixer (Genie 2, Scientific Industries). This was followed by addition of an equal volume of phenol-chloroform isoamyl alcohol (Sigma Aldrich). The samples were mixed thoroughly and sonicated (Branson 5800, 40 kHz, 20 °C) for 10 minutes. The samples were incubated at 65 °C for 10 minutes and centrifuged at 10,000 x g for 5 minutes at 20 °C (Centrifuge 5810, Eppendorf Austria GmbH). The aqueous DNA solution was mixed with an equal volume of HPLC-grade chloroform (≥99.8 %, Sigma-Aldrich) and centrifuged for 5 minutes at 10,000 x g at 20 °C, after which the organic phase was discarded. 2.5 volumes of -20 °C molecular grade ethanol (99 %, Sigma Aldrich) and 0.1 volumes of sterile 3 mM sodium acetate (pH 5.2, Merck KGaA) were added to one volume equivalent of DNA solution and incubated over-night at -20 °C. The precipitated DNA was centrifuged for 15 minutes at 14,000 x g and 4 °C. The pellet was carefully washed with 200 µL of ice-cold 70 % ethanol (Sigma Aldrich) prepared with ultra-pure water. A final centrifugation step was performed for 5 minutes at 14,000 x g and 4 °C, thereafter the supernatant was discarded, and the pellet was dried by vacuum centrifugation at 40 °C (centrifuge: VR-1, compressor: CT 110, both Heto Lab Equipment). The dry DNA pellets were dissolved in sterile, ultra-pure water (Invitrogen Nuclease Free Water, Thermo Fisher Scientific) and stored at -20 °C until further use.

Supplementary Note 7: NGS library preparation and sequencing

Targeted 16S cDNA libraries (active community) were constructed using the primer pair 515F and 806R, attached to a universal 5'tail for barcoding (Supplementary Table 10). PCR amplification was performed using the Platinum Taq polymerase (Thermo Scientific) and a dNTP mix (Kapa Biosystems, Sigma-Aldrich Co LLC) as instructed in the PCR setup guidelines with a MgCl₂ concentration of 4 mM. A template volume of 2.5 µL of DNA solution was used for each reaction mix. PCRs were performed with the following program: Initial denaturation: 3 min at 95 °C, 30 cycles: 20 s of denaturation at 95 °C, 30 s of annealing at 56 °C and 30 s of elongation at 72 °C; final elongation: 1 min at 72 °C. All reactions were performed in the peqSTAR Thermocycler 96 Universal Gradient (VWR). 16S rDNA libraries were constructed using the same protocol. Barcoding and sequencing described in this section were performed at Microsynth AG. To allow pooling of samples, the Nextera PCR libraries were barcoded with indexed adapters in a second PCR step. The Illumina MiSeq platform and a v2 nano 500 cycles kit was used to sequence PCR libraries. The produced paired-end reads that passed Illumina's chastity filter were subjected to de-multiplexing and trimming of Illumina adaptor residuals using Illumina's real-time analysis software (no further refinement or selection). The quality of the reads was checked with the software FastQC version 0.11.5⁴. The locus-specific V4 primers were trimmed from the sequencing reads with the software cutadapt v1.14⁵. Paired-end reads were discarded if the primer could not be trimmed.

Supplementary References:

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4. Andrews, S. FastQC: a quality control tool for high throughput sequence data. <https://www.bioinformatics.babraham.ac.uk/projects/fastqc/> (2010).
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